EXPERIMENT 7

DATA VISUALIZATION USING TABLEAU

Aim

To connect, analyze, visualize, and create interactive dashboards using Tableau for effective business intelligence and data analytics.

Software Requirements

• Tableau Desktop / Tableau Public

• Microsoft Excel

• CSV Dataset

• Windows Operating System

Theory

Tableau is a powerful Business Intelligence (BI) and Data Visualization tool used to create interactive dashboards and reports.

Tableau allows users to: Connect multiple data sources, create worksheets, build dashboards, create stories, perform data analysis, and share visual insights.

Tableau Architecture

Data Source ↓ Data Connection ↓ Worksheet ↓ Dashboard ↓ Story

Learning Outcomes

✓ Connect datasets

✓ Create worksheets

✓ Generate visualizations

✓ Build dashboards

✓ Create stories

✓ Use filters and parameters

✓ Analyze data interactively

Sample Dataset – Student Performance Dataset

Student	Department	Marks
Arun	CSE	85
Bala	IT	78
Divya	CSE	95
Priya	AI&DS	90
Ravi	IT	72
Karthik	AI&DS	88

The dataset contains 6 students from CSE, IT and AI&DS; departments. Marks range from 72 to 95.

Algorithm 1 – Connecting Data Source

Step 1 — Open Tableau Desktop

Step 2 — Select Connect

Step 3 — Choose Excel File / Text CSV

Step 4 — Browse Dataset

Step 5 — Open Dataset

Step 6 — Verify Imported Data

Procedure

Tableau ↓ Connect ↓ Microsoft Excel / Text CSV ↓ Student.xlsx / Dataset ↓ Open

Output

Student	Department	Marks
Arun	CSE	85
Bala	IT	78
Divya	CSE	95

Dataset loaded into Tableau successfully.

Algorithm 2 – Creating Worksheet

Step 1 — Open Sheet 1

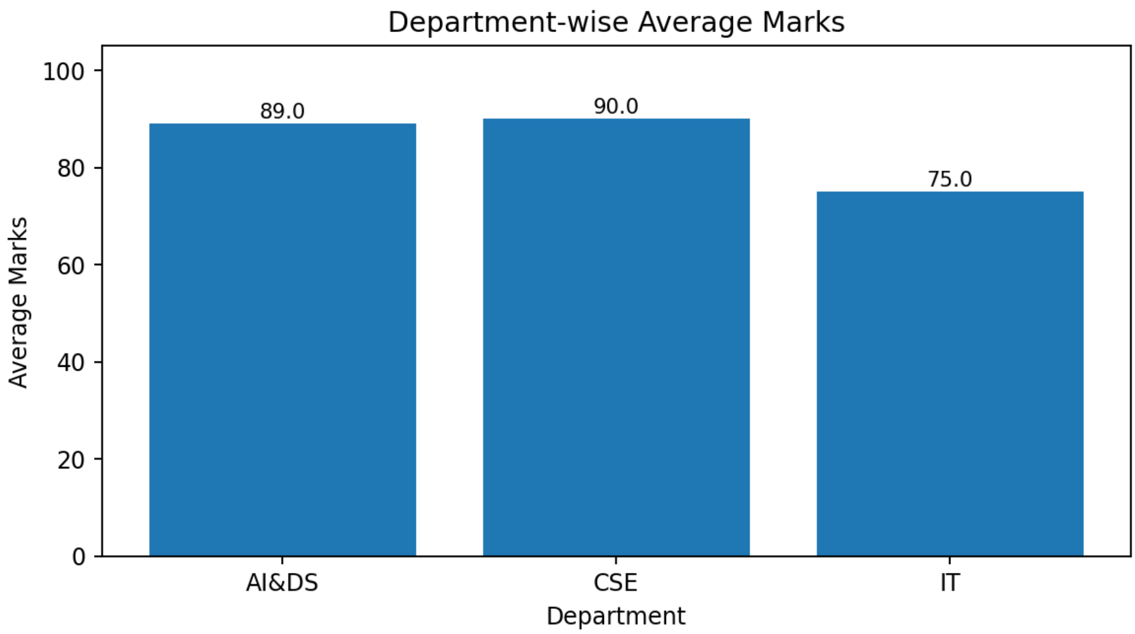
Step 2 — Drag Department to Columns

Step 3 — Drag Marks to Rows

Step 4 — Select Visualization

Step 5 — Analyze Results

Output Visualization



Worksheet created successfully. The department-wise visualization compares average marks.

Observation: CSE has the highest department average, followed by AI&DS; and IT.

Algorithm 3 – Bar Chart Creation

Step 1 — Drag Student → Columns

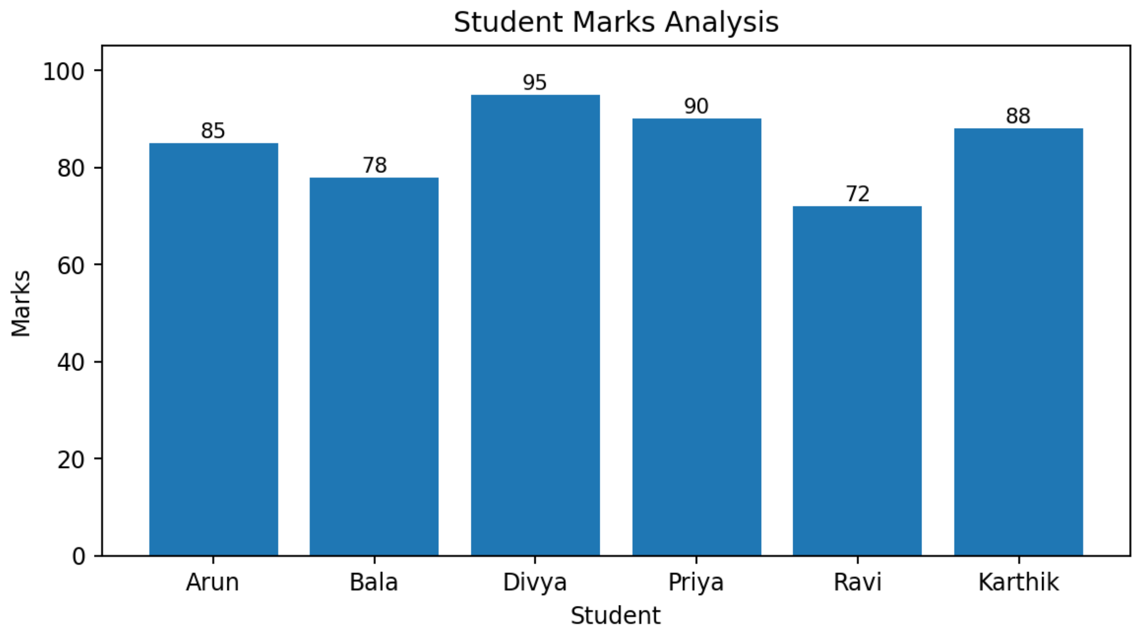
Step 2 — Drag Marks → Rows

Step 3 — Select Bar Chart

Step 4 — Apply Labels

Step 5 — Format Visualization

Output Visualization



Student Marks Analysis – Comparison of marks scored by students.

Interpretation: Highest Marks = Divya (95); Lowest Marks = Ravi (72).

Algorithm 4 – Pie Chart Creation

Step 1 — Drag Department

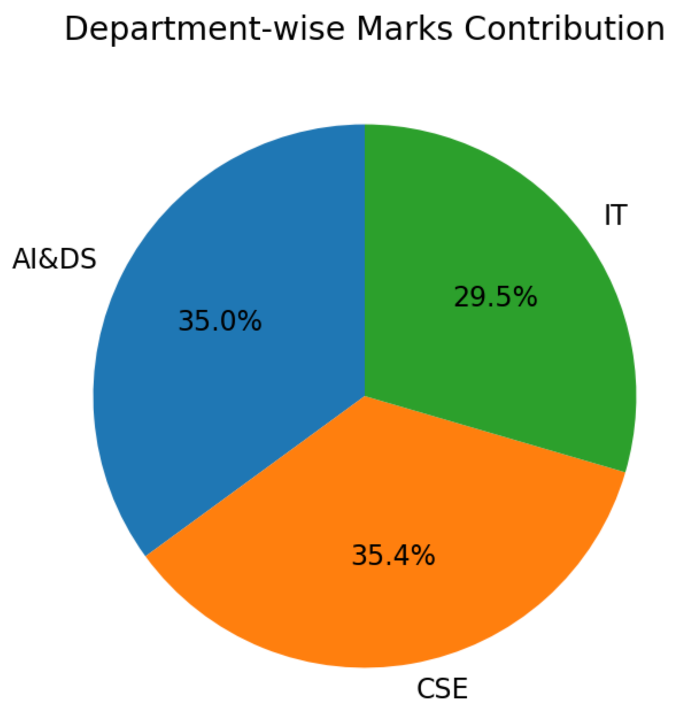
Step 2 — Drag Marks

Step 3 — Select Pie Chart

Step 4 — Show Labels

Step 5 — Analyze Distribution

Output Visualization



Department-wise Marks Contribution – Contribution of departments to total marks.

Interpretation: CSE contributes the highest total marks, while IT contributes the lowest.

Algorithm 5 – Line Chart Creation

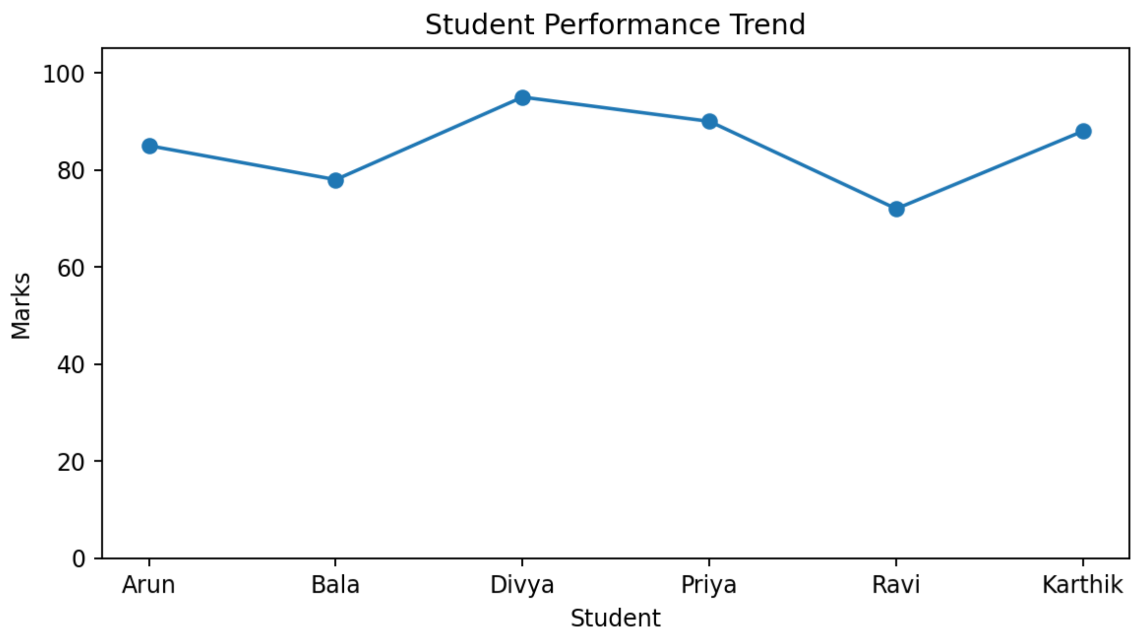
Step 1 — Drag Student

Step 2 — Drag Marks

Step 3 — Select Line Chart

Step 4 — Apply Formatting

Output Visualization



Student Performance Trend – Marks trend among students.

Algorithm 6 – Creating Calculated Field

Objective: Calculate Percentage Marks

Formula: (Marks / 100) × 100

Tableau expression: [Marks]

Procedure

Analysis → Create Calculated Field → Name: Percentage → Formula: [Marks]

Output Visualization

Student	Marks	Percentage
Arun	85	85%
Bala	78	78%
Divya	95	95%
Priya	90	90%
Ravi	72	72%
Karthik	88	88%

Algorithm 7 – Applying Filters

Step 1 — Drag Department to Filters

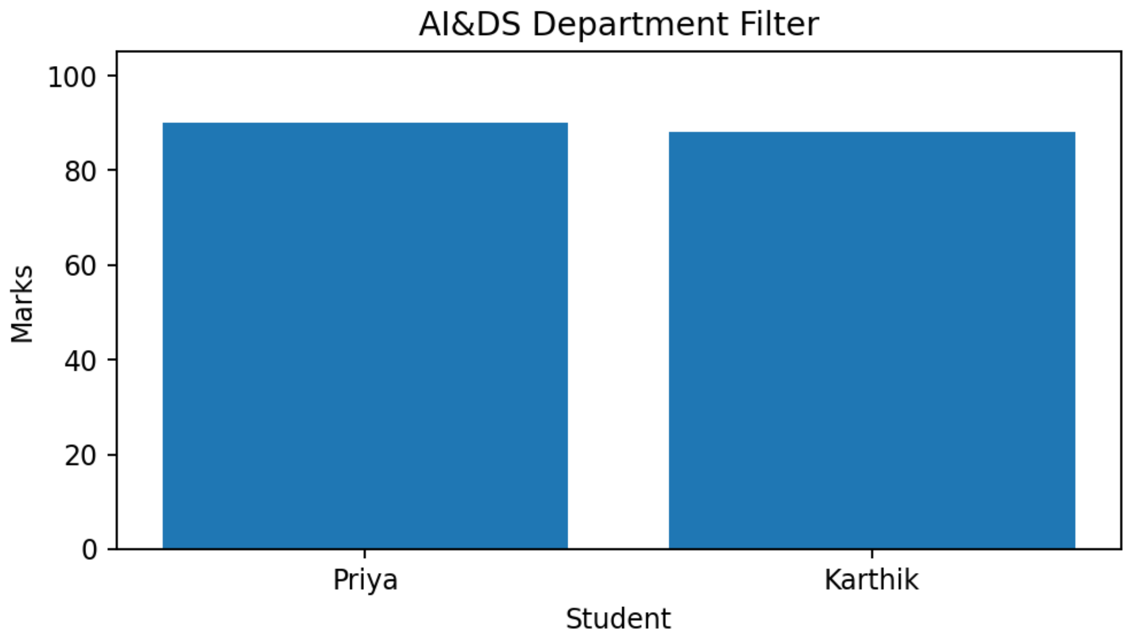
Step 2 — Select Department

Step 3 — Apply Filter

Step 4 — Observe Updated Charts

Example Filter: Department = AI&DS;

Output Visualization



Student	Department	Marks
Priya	AI&DS	90
Karthik	AI&DS	88

Algorithm 8 – Creating Parameter

Step 1 — Create Parameter

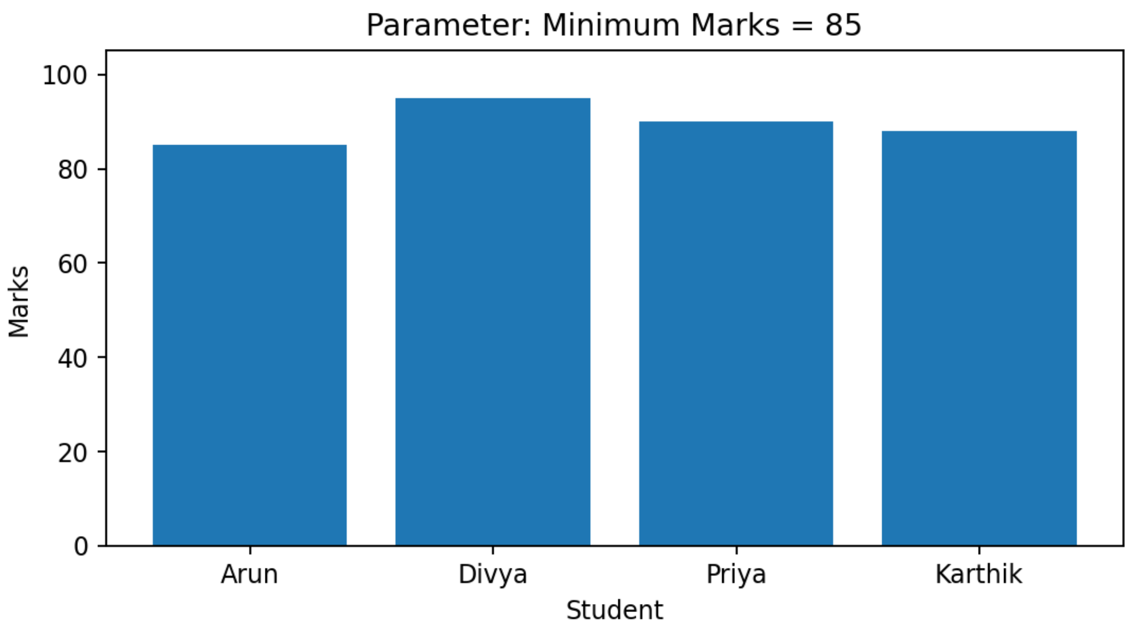
Step 2 — Specify Minimum Marks

Step 3 — Apply Parameter

Step 4 — Display Matching Records

Example: Minimum Marks = 85

Output Visualization



Student	Marks
Arun	85
Divya	95
Priya	90
Karthik	88

The parameter displays only records satisfying Marks ≥ 85.

Algorithm 9 – Dashboard Creation

Step 1 — Create Multiple Worksheets

Step 2 — Select Dashboard

Step 3 — Drag Worksheets

Step 4 — Arrange Components

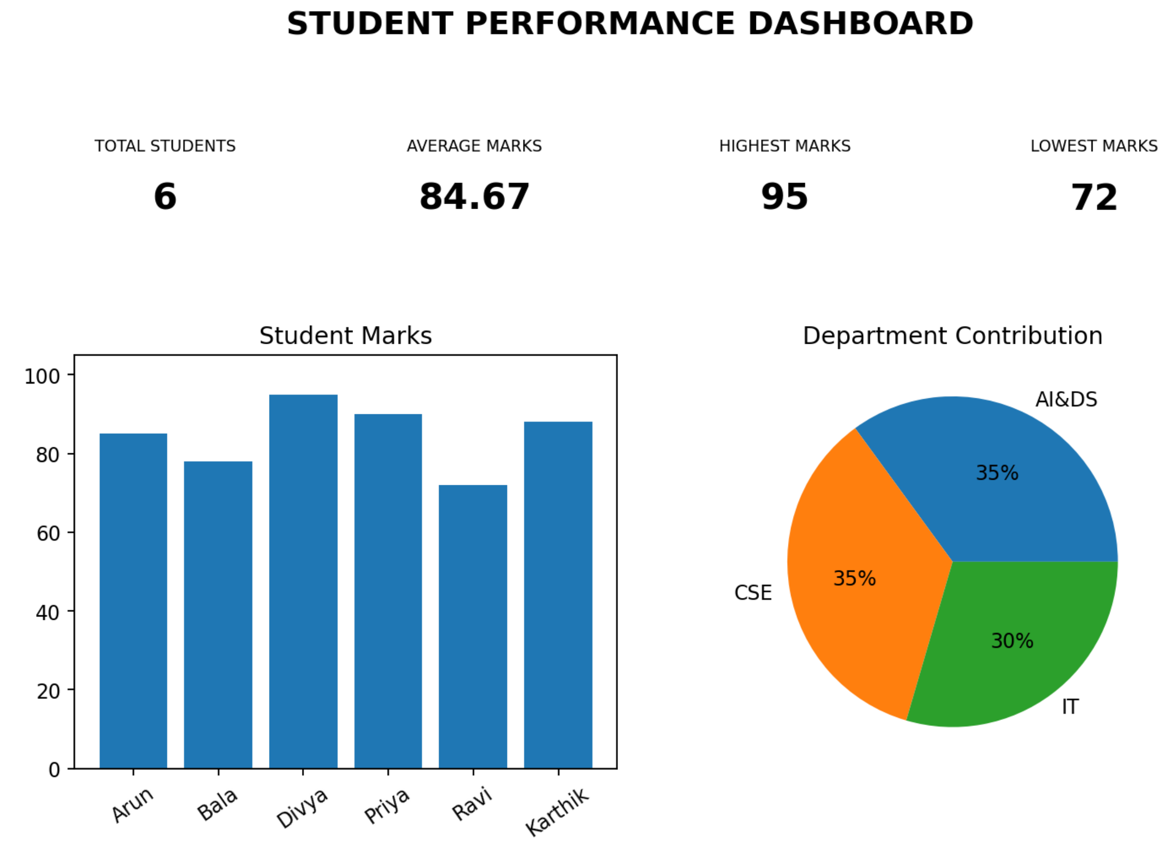
Step 5 — Add Filters

Step 6 — Save Dashboard

Dashboard Layout

Student Performance Dashboard → KPI Cards → Bar Chart + Pie Chart → Filter Panel

Output Visualization



Dashboard Analysis

KPI	Value
Total Students	6
Average Marks	84.67
Highest Marks	95
Lowest Marks	72

Dashboard Analysis: KPI cards provide a quick summary, while charts provide student-level and department-level comparisons. Filters allow interactive exploration.

Algorithm 10 – Story Creation

Step 1 — Create Dashboard

Step 2 — Select Story

Step 3 — Add Dashboard

Step 4 — Add Story Points

Step 5 — Present Insights

Example Story

Story Point	Insight
Story Point 1	Overall Student Performance
Story Point 2	Department-wise Analysis
Story Point 3	Top Performing Students
Story Point 4	Final Conclusions

Story Point 1: Overall performance using total students, average, highest and lowest marks.

Story Point 2: Department-wise comparison of CSE, IT and AI&DS.;

Story Point 3: Divya is the top-performing student with 95 marks.

Story Point 4: Interactive controls help present the findings clearly.

Applications

Domain	Application
Healthcare	Patient Analytics Dashboard
Banking	Loan Approval Analysis
Retail	Sales Dashboard
Education	Student Performance Dashboard

Result

The Student Performance dataset was successfully connected and analyzed using Tableau. Worksheets and multiple visualizations including bar charts, pie charts and line charts were generated. Calculated fields, filters and parameters were applied for interactive analysis. Finally, the visualizations were combined into a dashboard and organized into a story.

Conclusion

Tableau provides an effective platform for transforming tabular data into interactive visual insights. This experiment demonstrates the workflow from connecting a dataset to creating worksheets, applying analytical controls, designing a dashboard and presenting findings through a story.